

PrepPilot — Technology Stack

Senior project · Next.js 16 · FastAPI · 8 LLM models · 417 handlers · Sprint 5.7 (2026-05-18)

FRONTEND · web-app

deployed on Vercel

FRAMEWORK

Next.js 16

App Router · Server Components · Edge runtime
react 19 · typescript 5 strict

Tailwind CSS v4

@plugin "@tailwindcss/typography"
via @import "tailwindcss"

AUTH · ORM

NextAuth v5

Google OAuth · session cookies
PrismaAdapter

Prisma 6

schema-first · multi-provider
sqlite (dev) → mongo (prod)

UI · MARKDOWN · VIZ

react-markdown + remark-gfm

.chat-md · MD table → widget
tables · code · blockquotes

react-plotly.js

Dynamic import · SSR-safe · fullscreen
memoized · PrepPilot theme

FILE I/O · ICONS

SheetJS (xlsx)

20+ formats · csv/json/xlsx/yaml/xml
client-side parsing

@mui/icons-material

Outlined set · Stop/Reply/Send/etc
tree-shaken per icon

DATA · STORAGE

SQLite dev · MongoDB prod

RELATIONAL · DOCUMENT

SQLite

file:./prisma/dev.db
Local dev · NextAuth sessions
via Prisma

MongoDB Atlas

Production multi-tenant
Same Prisma schema · provider swap
MONGODB_URI env

SCHEMA · BLOB

Prisma ORM

Users · Sessions · Conversations
Dataset table · chat-local artifacts
prisma/schema.prisma

Object Store (Sprint 8)

dev: local fs (ml-datascience/models/)
prod: S3 / GCS for joblib + datasets
MODELS_DIR env

FILE FORMATS SUPPORTED (UPLOAD)

Text:

csv · tsv · json · jsonl · ndjson · yaml · yml · txt · md · xml · html

Spreadsheet:

xlsx · xls · xlsb · ods (SheetJS, client-side)

Columnar:

parquet · pq · feather · arrow (pyarrow, backend)

100 MB upload cap · client/backend split keeps proxy thin

BACKEND · ml-datascience

Render · Fly · Docker

CORE

FastAPI

async routes · Pydantic v2 models
uvicorn · Python 3.13

LangChain

langchain-anthropic + langchain-openai
cached factory · history truncate

DATAFRAME · NUMERIC

pandas · NumPy

DataFrame I/O · vectorized ops
core of every handler

pyarrow 22

parquet · feather · arrow
/parse-file route

ML · MODELING

scikit-learn · LightGBM · XGBoost

27 algorithms (12 clf · 11 reg · 4 cluster)
joblib persistence

Optuna · imbalanced-learn · Boruta

Hyperparam tuning · SMOTE/ADASYN
RFE feature selection

VIZ · SANDBOX

Plotly · seaborn · missingno

Interactive charts + matplotlib fallback
Thai font · PrepPilot theme

Sandboxed exec()

restricted ns · 30s timeout
_figure_has_data() guard

HOSTING · DEPLOYMENT

multi-region · autoDeploy on main

FRONTEND HOSTING

Vercel

Edge functions for API proxy · global CDN · preview deployments per PR
ML_BACKEND_URL · NEXTAUTH_URL · DATABASE_URL · MONGODB_URI

BACKEND HOSTING

Render (primary)

Docker · starter plan · Singapore
Persistent disk for joblib
render.yaml Blueprint

Fly.io (alternate)

Same Docker image · Singapore VM
auto_stop_machines · 1 GB RAM
fly.toml

CONTAINERS · LOCAL

Docker + Docker Compose

Single-command local boot: 'python run.py --docker' (installation-core/)
Dockerfile · docker-compose.yml · run.py launcher

Branch flow: feat/* → PR → main → autoDeploy (Render) · 'fly deploy' (Fly) · Vercel rebuilds frontend

CATEGORIES Frontend Backend AI / LLM Data · Storage Hosting Dev Tooling